

A Project Report On

GalactusAI — AI-Powered Resume Analyser

Submitted in partial fulfillment of the requirements for the degree

Under the Guidance of

INVERTIS UNIVERSITY

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PROJECT SPECIFICATION

1. Introduction

1.1 Purpose

GalactusAI is an AI-powered web application that analyses a user's resume and provides detailed feedback including ATS score, strengths, weaknesses, missing keywords, and suitable job roles.

1.2 Scope

- Users can upload their resume (PDF format)
- The system extracts text from the PDF accurately
- AI analyses the resume and returns a structured performance report
- Results are displayed in a visually appealing, interactive dashboard

1.3 Intended Users

- Students and freshers preparing for job applications
- Working professionals looking to improve their resume
- Anyone seeking ATS-optimized resume feedback

2. System Overview

The system operates on a modern client-server architecture. The user uploads a PDF via a React frontend. The request is sent to a backend processor where text extraction occurs. This data is then securely transmitted to an advanced Large Language Model (LLM) which evaluates the text against modern hiring standards.

3. Functional Requirements

ID	Requirement
FR-01	User can upload a PDF resume (max 5MB)
FR-02	System extracts text from the uploaded PDF using pdfplumber

FR-03	System sends extracted text to Groq AI (LLaMA 3.3 70B model)
FR-04	AI returns an Overall Score and an ATS Compatibility Score (0–100)
FR-05	AI returns Section-wise scores (Contact, Summary, Experience, Skills, Education)
FR-06	System returns strengths, missing keywords, and suggested job roles

4. Non-Functional Requirements

- **Performance:** Response time should be under 30 seconds.
- **Responsiveness:** Application should work flawlessly on both desktop and mobile browsers.
- **Security:** API keys must not be exposed on the frontend.

5. Tech Stack

Layer	Technology
Frontend	React 18, Vite, Tailwind CSS
Backend	Python 3, Flask, Flask-CORS
PDF Parsing	pdfplumber
AI Model	Groq API — LLaMA 3.3 70B Versatile
Hosting	Netlify (Frontend), Render (Backend)

6. Future Enhancements

- Support for .docx (Word) resume format parsing.
- Direct job description matching (paste a JD and get a match score).
- Capability to download the generated report as a PDF.
- User authentication system to save resume history and track improvements.